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# A unified viscoelastic-viscoplastic constitutive framework for optically clear adhesives: Experiments, modeling, and FE validation

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## Abstract

Optically clear adhesives (OCAs) are extensively employed as bonding layers in display manufacturing. As compliant interlayers in foldable display stacks, they undergo pronounced time and temperature -dependent deformation during repeated bending, which makes reliable constitutive characterization essential for accurate stress prediction and design optimization. Therefore, accurately characterizing their mechanical behavior is fundamental to bending-stress analysis and design optimization of foldable displays. Existing studies emphasize viscoelastic or hyperelastic behavior yet overlook the coupled viscoelastic-viscoplastic response. We propose a unified constitutive framework combining a generalized Maxwell representation with the Qian-Liu viscoplastic formulation to characterize OCA mechanics. The proposed model is calibrated against stress-relaxation experiments conducted at 30, 60, and 85 °C, from which the corresponding material parameters are identified; the fitting results demonstrate that the model captures the coupled viscoelastic-viscoplastic behavior under these conditions. The framework is further implemented into an ABAQUS UMAT and validated via finite-element simulations of a simplified foldable-display structure, which corroborate the experimental observations. The framework closes a gap in OCA constitutive modeling and delivers a calibrated, numerically validated tool for structural optimization and long-term reliability assessment of flexible OLED devices.

## Introduction

With the rapid advancement of display technologies, optically clear adhesives (OCA) have emerged as the preferred bonding material for transparent optical components owing to their outstanding mechanical properties, high optical transparency, strong adhesion capability, and long-term resistance to aging<sup>1–5</sup>. Flexible organic light-emitting diodes (OLEDs) and active-matrix OLEDs (AMOLEDs), characterized by high resolution, low power consumption, lightweight design, and wide viewing angles, play a pivotal role in the display industry. Their application scenarios continue to expand alongside the

development of personal smart terminals, attracting widespread attention and being regarded as one of the most promising directions in future display technologies<sup>6–11</sup>. Flexible OLED panels typically adopt a stacked architecture, in which the functional layers are bonded by a soft and transparent OCA, forming an alternating “hard-soft-hard-soft” multilayer structure<sup>12,13</sup>. Within such laminated systems, the soft adhesive layer serves as a critical component that often undergoes significant shear deformation during bending. This layer not only provides sufficient interfacial adhesion but also functions as a stress-buffering medium to accommodate bending-induced deformation<sup>14,15</sup>. Therefore, prior to OLED structural design and rapid reliability assessment, it is essential to comprehensively characterize the mechanical behavior of OCA and to establish a physically reasonable constitutive model<sup>16–18</sup>.

OCA is primarily synthesized through free-radical polymerization of acrylate monomers<sup>19</sup>, and it generally exhibits visco-hyperelastic behavior<sup>20</sup>. Numerous studies

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have investigated the constitutive modeling and finite element (FE) analysis of OCA. In 2014, Yeh et al. first employed FE methods to simulate the stress distribution in foldable OLED modules, analyzing the bending stresses in laminated foldable touch panels. In their study, OCA was simplified as a linear elastic material, without accounting for its inherent viscoelasticity or hyperelasticity<sup>21</sup>. Ma et al. determined the hyperelastic and viscoelastic properties of OCA through uniaxial tensile, shear, and stress relaxation tests to obtain material parameters, and subsequently proposed a three-dimensional FE bending model for the OLED bending region<sup>22</sup>. Zhang et al. further explored the hyperelastic and viscoelastic mechanical behavior of OCA by fitting experimental data with appropriate models, thereby elucidating the influence of temperature on its mechanical performance<sup>23</sup>. Salmon et al. conducted FE simulations to examine the impact of the viscoelastic constitutive properties of 3 M OCA on the interlayer mechanical behavior of foldable display panel stacks, highlighting its crucial role in enhancing foldability<sup>24</sup>. Zhao et al. considered the interplay between crosslinked networks, free chains, and entanglement constraints, successfully capturing the isotropic visco-hyperelastic behavior of OCA<sup>25</sup>. Cheng et al. employed several phenomenological hyperelastic models to characterize OCA mechanics and applied them to bending simulations of AMOLED displays<sup>26</sup>. Similarly, Jia et al. adopted a Yeoh hyperelastic framework to analyze bending stresses in foldable AMOLED panels incorporating OCA layers<sup>27</sup>. Other studies focusing on multilayer structures in foldable displays have utilized FE methods to investigate bending stress regulation, neutral-plane optimization, and delamination prevention, aiming to improve the structural reliability of OCA-bonded systems<sup>28,29</sup>. Collectively, these investigations demonstrate a gradual evolution from simplified linear elastic representations of OCA toward more advanced hyperelastic, viscoelastic, and visco-hyperelastic constitutive models. However, the viscoelastic-viscoplastic characteristics of OCA have not yet been considered.

Lee et al. observed that OCA exhibits permanent deformation during creep and stress relaxation measurements, which cannot be fully recovered upon stress removal<sup>30</sup>. Similarly, Park et al. reported the presence of unrecoverable residual strain in creep-recovery tests of OCA<sup>31</sup>, while Kim et al. demonstrated through strain recovery experiments that pristine adhesive films recover only 68% of the imposed strain, leaving approximately 32% as permanent, unrecoverable deformation<sup>32</sup>. These phenomena are strongly aligned with the fundamental characteristics of viscoplastic materials. According to established theoretical frameworks, the total strain in viscoplastic materials can be decomposed into elastic and irreversible plastic components, wherein the irreversibility

of plastic strain directly corresponds to the permanent deformation observed in OCA. The plastic strain rate is typically governed by a Norton-type relation with stress, reflecting both time and loading-rate dependencies—features that are consistent with the time-dependent permanent deformation of OCA. Despite the existence of partial static recovery, the pronounced residual strain in OCA unequivocally confirms the presence of plastic deformation, representing a prototypical viscoplastic response<sup>33</sup>. Li et al. further proposed that the constitutive relationship of OCA can be categorized into three regimes: (i) the stress-strain response under rapid loading, (ii) the creep constitutive relation under sustained loading, and (iii) the stress-strain response during rapid unloading<sup>34</sup>. Therefore, investigating the viscoelastic-viscoplastic constitutive behavior of OCA is of profound importance for a comprehensive understanding of its mechanical performance.

To date, no studies have reported the development of a viscoelastic-viscoplastic constitutive model specifically for OCA. However, significant progress has been achieved in modeling other classes of adhesives. For example, Chen et al. proposed a one-dimensional nonlinear viscoelastic-viscoplastic model that couples the Schapery nonlinear single-integral formulation with the Perzyna viscoplastic framework, which successfully captured the mechanical behavior of adhesives<sup>35</sup>. Groth employed three different viscoelastic-viscoplastic models to analyze stress states in single-lap adhesive joints with and without cracks<sup>36</sup>. Liang et al. combined a multi-relaxation viscoelastic-viscoplastic formulation with a three-element viscoelastic model to construct a constitutive framework capable of describing the mechanical evolution of adhesives<sup>37</sup>. Qi et al. developed an elastoviscoplastic model based on a rheological system composed of a Voigt element in series with a spring and in parallel with multiple Maxwell elements, effectively characterizing creep, relaxation, and stress-strain responses<sup>38</sup>. Collectively, these studies provide a solid theoretical foundation, feasible modeling strategies, and robust validation methods that may guide the future development of a viscoelastic-viscoplastic constitutive model for OCA.

In this context, the present work focuses on the coupled viscoelastic-viscoplastic mechanical behavior of OCA, aiming to establish a unified constitutive framework by integrating the generalized Maxwell viscoelastic model with the Qian-Liu viscoplastic formulation. Key parameters are extracted from stress relaxation experiments conducted under multiple temperature conditions, and the proposed model is validated through nonlinear regression and finite element simulations. This systematic approach enables the elucidation of temperature and time-dependent mechanisms governing the mechanical response of OCA. Beyond advancing the theoretical

understanding, the proposed methodology provides accurate predictive tools for structural optimization and long-term reliability evaluation of flexible OLED devices, while simultaneously filling a critical gap in the field of viscoelastic-viscoplastic constitutive modeling.

## Constitutive model

### Viscoelastic Constitutive Model

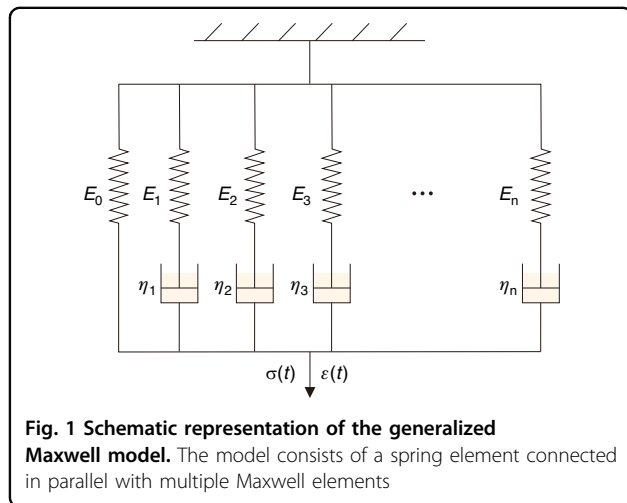
OCA exhibits typical viscoelastic mechanical behaviors such as stress relaxation and creep, which can be effectively characterized by establishing an appropriate viscoelastic constitutive model. Commonly used viscoelastic models in related studies include the Maxwell model, the generalized Maxwell model, the Kelvin model, and various four-parameter models. To accurately describe the viscoelastic response of OCA, this study adopts Hookean springs and dashpots as the fundamental rheological elements, representing ideal elastic solids and viscous fluids, respectively. The static and dynamic viscoelastic responses of the material were systematically examined, and the generalized Maxwell model was selected to analyze the viscoelastic behavior of OCA.

As illustrated in Fig. 1, the generalized Maxwell model consists of several Maxwell elements arranged in parallel with an additional spring to capture relaxation behavior. Under constant applied strain, the stress response of a basic Maxwell element can be expressed as<sup>39</sup>:

$$\sigma(t) = \varepsilon_0 E e^{-\frac{t}{\tau}} \quad (1)$$

Where  $E$  is the elastic modulus of the spring,  $\varepsilon_0$  is the applied strain, and  $\tau = \frac{\eta}{E}$  represents the relaxation time constant of the element.

Since all units connected in parallel undergo the same strain while the total stress is the summation of the stresses carried by each unit, the stress-strain relationship of the generalized Maxwell model can be expressed as



follows:

$$\begin{aligned} \sigma &= \sigma_1 + \sigma_2 + \dots + \sigma_n \\ \varepsilon &= \varepsilon_1 = \varepsilon_2 = \dots = \varepsilon_n \end{aligned} \quad (2)$$

The governing differential equation for the  $i$ -th Maxwell element can be expressed as:

$$\frac{d\varepsilon}{dt} = \frac{1}{E_i} \frac{d\sigma_i}{dt} + \frac{\sigma_i}{\eta_i} \quad (3)$$

The stress relaxation modulus can be expressed as:

$$E(t) = E_\infty + \sum_{i=1}^N E_i \cdot e^{-\frac{t}{\tau_i}} \quad (4)$$

Where  $E(t)$  denotes the relaxation modulus of the generalized Maxwell model;  $E_\infty$  is the long term modulus and  $E_i$  represents the  $i$ -th Maxwell element. The normalized modulus  $g_i$  is frequently employed as a dimensionless descriptor of material constants, and together with the relaxation time  $\tau_i$  constitutes the Prony series, which is widely used for defining viscoelastic properties and for numerical implementation. The expression for  $g_i$  is given by:

$$g(t) = 1 - \sum_{i=1}^N g_i \left(1 - e^{-\frac{t}{\tau_i}}\right) \quad (5)$$

According to the Boltzmann superposition principle, the stress response of the generalized Maxwell model can be expressed as:

$$\sigma(t) = \varepsilon_0 E(t) + \int_0^t E(t-\xi) \frac{d\varepsilon(\xi)}{d\xi} d\xi \quad (6)$$

In the subsequent experiments, OCA will be subjected to stress-relaxation and dynamic viscoelastic tests, and the resulting data will be interpreted using the generalized Maxwell framework to identify the parameters pertinent to the viscoelastic constitutive model.

### Viscoplastic Constitutive Model

In this study, we adopt the Qian-Liu model to characterize the viscoplastic behavior of OCA. As a damage-coupled unified viscoplastic constitutive framework grounded in irreversible thermodynamics, it enables the coupling of elastic and viscoplastic damage and effectively captures time- and rate-dependent viscoplastic deformation, complex loading histories, and creep-fatigue interactions; it is therefore well suited for describing the viscoplastic response of OCA<sup>17,40,41</sup>.

The general form of the Qian-Liu viscoplastic model is given by<sup>17</sup>:

$$\dot{\epsilon}_{ss}^I = \Theta(T) [\sinh(A\sigma_{ss})]^n \quad (7)$$

$$\dot{\epsilon}^I = \Theta(T) \left[ \sinh\left(\frac{\sigma - X_1 - X_2}{D}\right) \right]^n \quad (8)$$

$$X_1 = \frac{A\sigma_{ss}}{A_1} \left[ 1 - \exp\left(-\frac{A_1 H_1}{A\sigma_{ss}} \epsilon^I\right) \right] \quad (9)$$

$$X_2 = \frac{A\sigma_{ss}}{A_2} \left[ 1 - \exp\left(-\frac{A_2 H_2}{A\sigma_{ss}} \epsilon^I\right) \right] \quad (10)$$

Where  $\Theta(T)$  denotes the thermal activation term, whose explicit expression is:

$$\Theta(T) = \dot{\epsilon}_0 \exp\left(-\frac{Q}{kT}\right) \quad (11)$$

In the formulation, the back stresses  $X_1$  and  $X_2$  are introduced to capture the short and long-range components of dislocation-mediated internal stress;  $\dot{\epsilon}^I$  and  $\epsilon^I$  denote the inelastic strain rate and inelastic strain, respectively;  $Q$  is the apparent activation energy;  $T$  the absolute temperature;  $G$  the shear modulus;  $D$  the drag stress;  $\sigma_{ss}$  the saturation stress;  $H_1$  and  $H_2$  are the strain-hardening coefficients, and these parameters will change with temperature, affecting the strain; and  $A$ ,  $A_1$ ,  $A_2$  are the back-stress coefficients. The Qian-Liu model conforms to the canonical Garofalo-Arrhenius form and employs two back-stress components to represent short and long-range internal stress effects, thereby providing a micromechanically grounded (dislocation-based) explanation of how temperature and strain rate govern the material's mechanical response.

Given the rapid stress saturation under tensile loading, the plastic strain rates in Eqs. (7) and (8) can be regarded as equivalent, yielding the following expression:

$$\sigma = \text{Darcsinh} \left[ \frac{\dot{\epsilon}_{ss}^I}{\Theta(T)} \right]^{\frac{1}{n}} + X_1 + X_2 \quad (12)$$

For comparison, He et al. proposed a modified Qian-Liu formulation by introducing an additional temperature-dependent hardening stress term, which improves the description of temperature-dependent behavior. The modified expression can be written as:

$$\sigma = \text{Darcsinh} \left[ \frac{\dot{\epsilon}_{ss}^I}{\Theta(T)} \right]^{\frac{1}{n}} + X_1 + X_2 + \sigma_H(T) \quad (13)$$

To facilitate subsequent parameter identification, we derive the differential form of the Qian-Liu model

and obtain the time-dependent relation for its tangent modulus.

Starting from:

$$\sigma = E\epsilon^I \quad (14)$$

Differentiating both sides of Eq. (13) with respect to time yields:

$$\frac{d\sigma}{dt} = \frac{dE}{dt} \epsilon^I + E(t) \frac{d\epsilon^I}{dt} \quad (15)$$

Here, we simplify the model by assuming that:

$$\epsilon^I = kt \quad (16)$$

Where  $k$  denotes the strain rate. This assumption is based on the steady-state creep behavior, where the plastic strain rate becomes constant after the initial transient phase.

By substituting Eqs. (12) and (15) into Eq. (14) and integrating both sides, the time-dependent relation for the tangent modulus of the Qian-Liu model is obtained as:

$$E(t) = \frac{A\sigma_{ss}}{kt} \left( \frac{1 - e^{-\frac{A_1 H_1 k}{A\sigma_{ss}} t}}{A_1} + \frac{1 - e^{-\frac{A_2 H_2 k}{A\sigma_{ss}} t}}{A_2} \right) \quad (17)$$

### Viscoelastic-Viscoplastic Constitutive Model

In this study, the total strain is decomposed into viscoelastic and viscoplastic components, which are assumed to be uncoupled<sup>35,42</sup>:

$$\epsilon^{\text{total}} = \epsilon^{\text{ve}} + \epsilon^{\text{vp}} \quad (18)$$

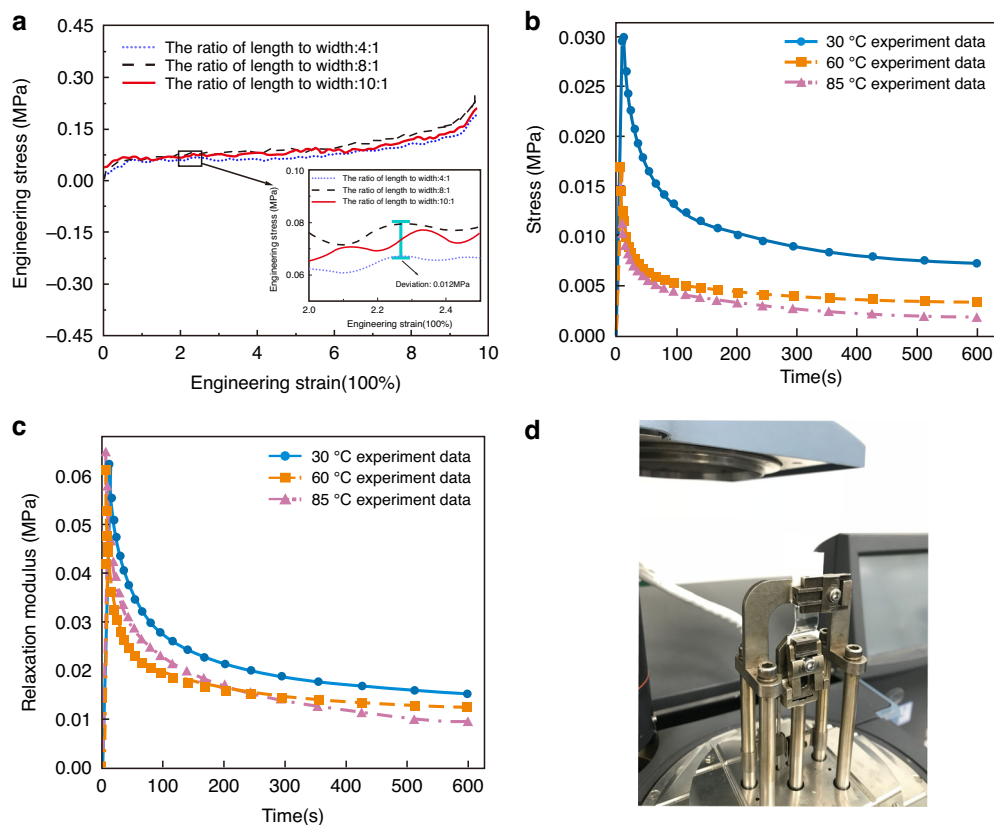
where the superscripts total, ve, and vp denote the total strain and the viscoelastic and viscoplastic strain components, respectively. By multiplying Eqs. (4) and (16) by their corresponding strains and then summing, the viscoelastic-viscoplastic constitutive relation for OCA is obtained.

$$\sigma(t) = \left[ E_{\infty} + \sum_{i=1}^N E_i e^{-\frac{t}{\tau_i}} \right] \epsilon^{\text{ve}} + \frac{A\sigma_{ss}}{kt} \left( \frac{1 - e^{-\frac{A_1 H_1 k}{A\sigma_{ss}} t}}{A_1} + \frac{1 - e^{-\frac{A_2 H_2 k}{A\sigma_{ss}} t}}{A_2} \right) \epsilon^{\text{vp}} \quad (19)$$

## Materials and Methods

### Specimen Selection

At present, there is no unified standard for the preparation of OCA specimens. Zhang et al. conducted uniaxial tensile tests at a constant tensile rate using specimens with different aspect ratios and found that, at



**Fig. 2** Experimental setup and primary mechanical testing data. **a** Stress-strain data for specimens with different aspect ratios<sup>23</sup>; **b** stress and **c** relaxation modulus as a function of time from stress-relaxation experiments; **d** specimen maintained in tensile mode during testing

a fixed rate of 100 mm/min and a temperature of 30 °C, both the scatter and the trends of the measured data were essentially identical across aspect ratios; the size-induced deviation was within 10%, as shown in Fig. 2(a), indicating that the material response is effectively scale-independent<sup>23</sup>. Accordingly, to meet the requirements of the testing apparatus, specimens with dimensions 40 mm×10 mm×1 mm were employed for subsequent experiments.

### Experimental Procedure

Tests of static mechanical behavior were conducted primarily via stress-relaxation and creep experiments, both of which probe the time-dependent response of the material. In stress-relaxation measurements, polymer specimens are subjected to an instantaneous deformation at a prescribed temperature, and the decay of the stress required to maintain the constant strain is recorded; the imposed strain is a step input. Stress-relaxation testing is particularly advantageous for elucidating polymer mechanics from the perspective of molecular mobility. Moreover, temperature exerts a pronounced influence on the mechanical response of OCA—manifested in

differences in relaxation times and equilibrium moduli—so examining stress-relaxation behavior over a range of temperatures facilitates a micro-to-macro understanding of its macroscopic mechanical characteristics.

Uniaxial stress-relaxation tests at 30, 60, and 85 °C were conducted on three specimens per group, and dynamic viscoelastic tests were performed on the OCA material; these temperatures were selected to span near-ambient, intermediate, and engineering-relevant high-temperature conditions commonly adopted in optoelectronic reliability assessment. For all three temperatures, the same target tensile strain of 50% was applied and held constant for 10 min, ensuring consistent strain conditions for cross-temperature comparison. The resulting time evolutions of stress and relaxation modulus are presented in Fig. 2b, c and were subsequently used to identify parameters for the viscoelastic and viscoelastic-viscoplastic constitutive models. Figure 2d shows the specimen, which remained in tensile mode throughout the experiment.

The dynamic viscoelastic tests were carried out using the DMA Q800, where a 0.01 N preload force and a 125% force track were applied. The force track adjusted the total force to avoid the sample from automatically buckling

during the test. The amplitude of the sinusoidal force applied to the samples was set to  $20\mu\text{m}$ . Dynamic mechanical analysis (DMA) was performed to investigate the temperature-dependent viscoelastic behavior of the OCA material. Temperature sweeps were conducted over a broad range of temperatures, from  $-40^\circ\text{C}$  to  $125^\circ\text{C}$ , with oscillation frequencies selected at 1 Hz, 2 Hz, 5 Hz, and 10 Hz. The temperature was changed at a rate of  $3^\circ\text{C}$  per minute, and this range was selected to refer to the J-STD-020D standards. The resulting data were processed to establish the viscoelastic model.

## Results and discussions

This section first performs parameter identification from the stress-relaxation data using both a viscoelastic constitutive description and a viscoelastic-viscoplastic counterpart. The viscoelastic response is represented by a generalized Maxwell model, whereas the viscoplastic response is formulated within the Qian-Liu framework via its evolution equations. During fitting, to ensure cross-temperature comparability and numerical stability, the strain-rate parameter was held constant. Material parameters for each model were obtained by nonlinear least-squares regression, followed by a quantitative comparison to assess the suitability and accuracy of the two models in capturing the stress-relaxation behavior. The constitutive models developed in Section 2 were subsequently implemented in ABAQUS via a UMAT user subroutine to construct a finite-element model of a flexible display and conduct numerical simulations, after which the simulation results were validated and analyzed.

### Parameter Identification for the Viscoelastic Constitutive Model

To quantitatively identify the temperature-dependent viscoelastic characteristics of OCA, this section calibrates the constitutive parameters using the stress-relaxation and DMA datasets described in Experimental Procedure. Specifically, the relaxation response is mapped to the constitutive formulation through least-squares minimization between the measured and predicted stress histories, from which the viscoelastic and viscoelastic-viscoplastic parameters at 30, 60, and  $85^\circ\text{C}$  are determined.

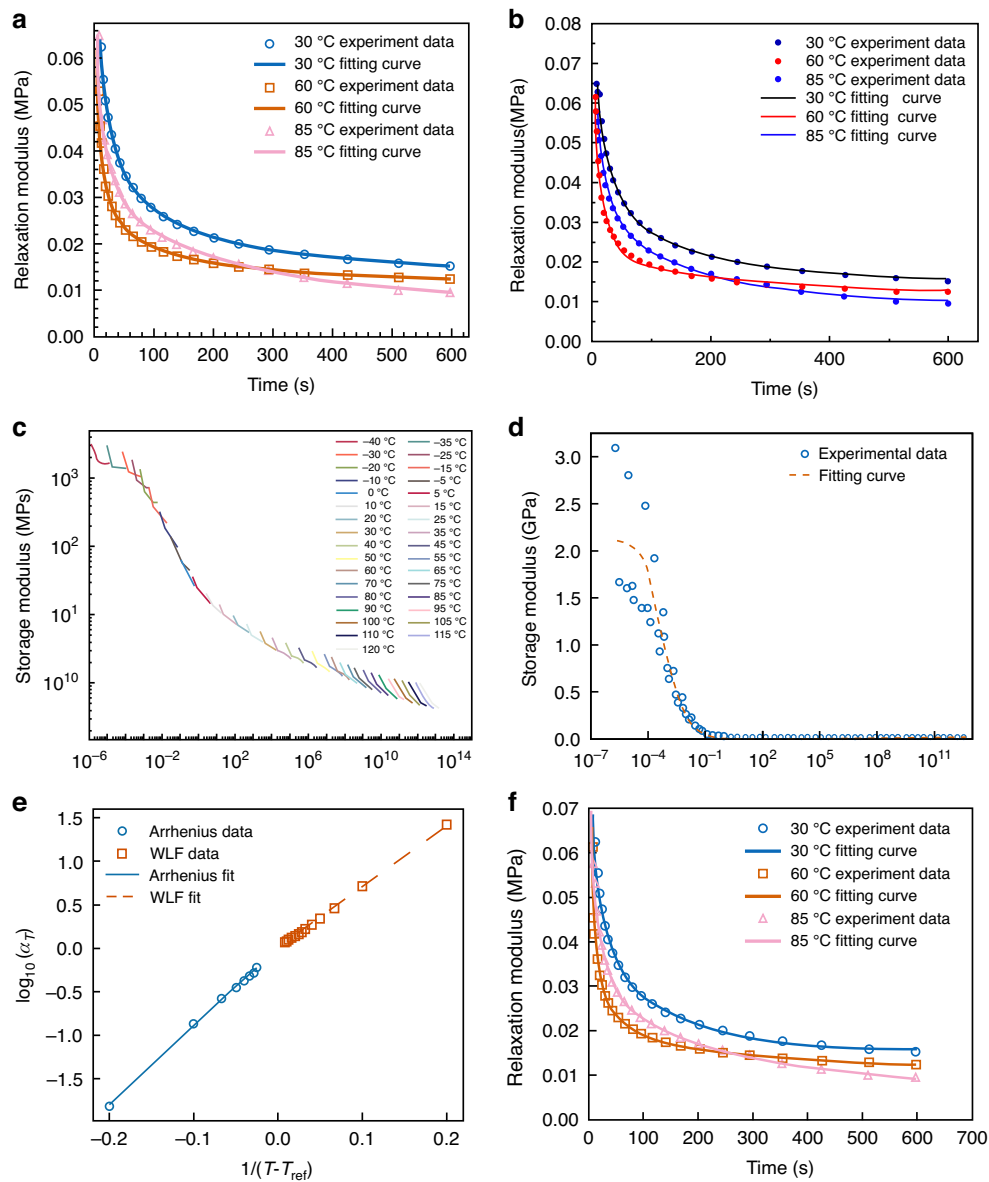
As shown in Fig. 2b, c, both the stress and the relaxation modulus rise sharply within a very short time before the target strain is reached; upon entering the relaxation stage, both decrease markedly with time and eventually approach steady values. At  $85^\circ\text{C}$ , the peak stress is approximately 12.65 kPa and rapidly decays to an equilibrium value of 1.84 kPa; correspondingly, the relaxation modulus decreases from 64.80 kPa to 9.51 kPa. These observations underscore the pronounced influence of temperature on viscoelastic behavior: increasing

temperature enhances segmental thermal mobility, enlarges the free volume, and reduces intermolecular friction, thereby accelerating stress relaxation and lowering both the equilibrium stress and the equilibrium relaxation modulus. At low temperatures, stress decay is primarily constrained by local segmental motions within the crosslinked network, whereas at elevated temperatures cooperative segmental dynamics and network rearrangement are significantly intensified, leading to a much faster relaxation process.

To establish a constitutive relation suitable for subsequent finite-element analysis, the relaxation responses at each temperature were fitted using a third-order generalized Maxwell model so as to capture viscoelastic features across multiple time scales; goodness of fit was evaluated by the coefficient of determination. As shown in Fig. 3a, with the corresponding Prony coefficients listed in Table 1, the fitted curves exhibit excellent agreement with the experimental data, with  $R^2$  values exceeding 0.98, indicating that the model reproduces the stress-relaxation behavior of OCA with high fidelity. The parameter trends show that the three relaxation time constants  $\tau_1, \tau_2, \tau_3$  correspond to fast segmental motion, mesoscale cooperative segmental dynamics, and long-time structural relaxation, respectively. With increasing temperature, all three mechanisms accelerate markedly, leading to pronounced reductions in their characteristic relaxation times  $\tau_i$ . Meanwhile, the modal relaxation modulus  $g_i$  decrease, reflecting an overall reduction in stiffness across relaxation modes. This trend is consistent with prior findings for acrylate-based structural adhesives and underscores that the temperature sensitivity of OCA's viscoelasticity is closely linked to polymer chain flexibility.

To assess the model's external validity, a comparison was conducted with the generalized Maxwell fit reported in Ref. 23, as both studies used the same experimental conditions and sample material. As shown in Fig. 3b and detailed in Table 1, both models successfully capture the temperature-dependent viscoelastic behavior of OCA. However, at  $60^\circ\text{C}$ , the second-order model used in Ref. 23 shows notable deviations in the intermediate time range. In contrast, the third-order model employed here—using the same dataset—achieves significantly improved accuracy, with the coefficient of determination increasing from  $R^2 > 0.97$  to  $R^2 = 0.9998$ . This confirms the superior capability of the third-order formulation in resolving multi-scale relaxation dynamics, particularly in capturing mid-term transitions.

To further refine the quantitative characterization of OCA's viscoelasticity, we complemented the analysis with dynamic mechanical testing. The storage-modulus data obtained from the temperature sweeps were processed using the time-temperature superposition (TTS) principle. The modulus curves measured at different



**Fig. 3 Constitutive model fitting and parameter identification.** **a** Generalized Maxwell model fits at different temperatures; **b** Benchmark: generalized Maxwell fitting results from Ref. <sup>23</sup>; **c** Storage-modulus master curve at the reference temperature of 0 °C; **d** Generalized Maxwell model fit to the master curve; **e** Master-curve shift factors and corresponding WLF/Arrhenius fits; **f** Viscoelastic-viscoplastic constitutive model fits at different temperatures

temperatures were horizontally shifted along the logarithmic time axis to construct a storage-modulus master curve at a reference temperature of 0 °C, as shown in Fig. 3c. The fitting results indicate that the model accurately captures the viscoelastic behavior across a wide range of temperatures, providing a reliable basis for further analysis. The master curve was then fitted using a third-order generalized Maxwell model, enabling faithful representation of the viscoelastic response over extended time scales. As shown in Fig. 3d, the fitted master curve

provides a clear depiction of the viscoelastic behavior across various temperatures, further supporting the validity of the model. The corresponding parameters for the third-order Prony series are as follows:  $g_1 = 0.845$ ,  $\tau_1 = 2.18 \times 10^{-4}$ ;  $g_2 = 0.911$ ,  $\tau_{2i} = 1.69 \times 10^{-3}$ ; and  $g_3 = 0.354$ ,  $\tau_4 = 3.68 \times 10^{-2}$ . Additionally, the analysis of the shift factors obtained during master-curve construction confirms the effectiveness of the time-temperature superposition principle in capturing the material's behavior at different temperatures. Fitting of the shift factors

obtained during master-curve construction revealed that the high-temperature regime ( $T > T_g$ , where  $T_g$  refers to the glass transition temperature of the material) follows the Williams-Landel-Ferry (WLF) equation, whereas the low-temperature regime ( $T < T_g$ ) conforms to an Arrhenius relation, as follows:

$$\frac{1}{\log_{10}^{\alpha_T}} = \frac{R \cdot \log_{10}^{\delta H}}{\delta H} \cdot \left( \frac{1}{T + 273} - \frac{1}{T_{\text{ref}} + 273} \right)^{-1} \quad (20)$$

$$= k \cdot \frac{1}{T} + b \quad T < 273\text{K}$$

$$\log_{10}^{\alpha_T} = -\frac{C_1(T - T_{\text{ref}})}{C_2 + (T - T_{\text{ref}})} \quad T > 273\text{K} \quad (21)$$

**Table 1 Comparison of generalized Maxwell Prony coefficients across temperature—this work vs. Ref. <sup>23</sup>**

| Temperature | i | $g_i$ in this work | $g_i$ in Ref. <sup>23</sup> | $\tau_i$ in this work | $\tau_i$ in Ref. <sup>23</sup> |
|-------------|---|--------------------|-----------------------------|-----------------------|--------------------------------|
| 30°C        | 1 | 0.4590             | 0.4248                      | 15.115                | 14.368                         |
|             | 2 | 0.2371             | 0.2436                      | 66.512                | 37.209                         |
|             | 3 | 0.1538             | 0.3315                      | 338.661               | 161.71                         |
| 60°C        | 1 | 0.6780             | 0.8166                      | 4.608                 | 15.034                         |
|             | 2 | 0.1504             | 0.1666                      | 24.992                | 203.425                        |
|             | 3 | 0.0839             | -                           | 184.486               | -                              |
| 85°C        | 1 | 0.5282             | 0.6172                      | 4.761                 | 16.281                         |
|             | 2 | 0.2293             | 0.2256                      | 25.734                | 118.25                         |
|             | 3 | 0.1799             | 0.1563                      | 241.154               | 236.26                         |

In these expressions,  $C_1$  and  $C_2$  are the universal constants of the WLF equation; the reference temperature is set to  $T_{\text{ref}} = 0^\circ\text{C}$ ; and  $\delta H$  denotes the activation energy. The fitting results for the two relations are presented in Fig. 3e. The corresponding parameters for the WLF equation are  $C_1 = -314.2019$  and  $C_2 = 43.9834$ , while the parameters for the Arrhenius equation are  $k = 8.9812$  and  $b = -0.0023064$ . They correspond, respectively, to relaxation mechanisms dominated by segmental motion and by small-molecule motion, consistent with the temperature dependence of chain mobility observed in the stress-relaxation experiments.

The mutual corroboration between dynamic and static viscoelastic tests provides multi-dimensional parameter support for constructing the OCA viscoelastic constitutive model, ensuring accurate representation of the material's mechanical response across broad temperature and time scales.

#### Parameter Identification for the Viscoelastic-Viscoplastic Constitutive Model

Building on a combined generalized Maxwell viscoelastic model and the Qian-Liu viscoplastic framework, we performed parameter identification from stress-relaxation experiments conducted at 30, 60, and 85 °C. The fitted responses are shown in Fig. 3f, and the identified parameters are summarized in Table 2. By incorporating back-stress and strain-hardening terms, the model systematically elucidates the temperature-dependent mechanisms governing the material response and enables comparison and validation against established frameworks in the literature.

In the viscoelastic part, the fittings indicate that the instantaneous modulus and the modal relaxation moduli

**Table 2 Identified parameters for the viscoelastic-viscoplastic constitutive model**

| Parameter category      | Parameter  | 30°C                        | 60°C                       | 85°C                       |
|-------------------------|------------|-----------------------------|----------------------------|----------------------------|
| Viscoplastic parameters | $A$        | $0.000217 \text{ MPa}^{-1}$ | $32.357 \text{ MPa}^{-1}$  | $55.2781 \text{ MPa}^{-1}$ |
|                         | $A_1$      | $992.295 \text{ MPa}^{-1}$  | $815.396 \text{ MPa}^{-1}$ | $805.166 \text{ MPa}^{-1}$ |
|                         | $A_2$      | $992.282 \text{ MPa}^{-1}$  | $813.468 \text{ MPa}^{-1}$ | $804.600 \text{ MPa}^{-1}$ |
|                         | $H_1$      | $23.1466 \text{ MPa}$       | $393.812 \text{ MPa}$      | $139.369 \text{ MPa}$      |
|                         | $H_2$      | $79.796 \text{ MPa}$        | $272.154 \text{ MPa}$      | $205.569 \text{ MPa}$      |
|                         | $E_\infty$ | $0.0153 \text{ MPa}$        | $0.0116 \text{ MPa}$       | $0.0068 \text{ MPa}$       |
| Viscoelastic parameters | $E_1$      | $0.0185 \text{ MPa}$        | $0.0459 \text{ MPa}$       | $0.0079 \text{ MPa}$       |
|                         | $E_2$      | $0.0449 \text{ MPa}$        | $0.0102 \text{ MPa}$       | $0.0138 \text{ MPa}$       |
|                         | $E_3$      | $0.0241 \text{ MPa}$        | $0.0088 \text{ MPa}$       | $0.0192 \text{ MPa}$       |
|                         | $\tau_1$   | $0.415 \text{ S}$           | $5.060 \text{ S}$          | $7.859 \text{ S}$          |
|                         | $\tau_2$   | $18.451 \text{ S}$          | $34.119 \text{ S}$         | $32.510 \text{ S}$         |
|                         | $\tau_3$   | $146.756 \text{ S}$         | $224.995 \text{ S}$        | $276.661 \text{ S}$        |

decrease progressively with increasing temperature, whereas the relaxation times shift toward longer time scales. This trend suggests that, at low temperatures, chain segments are relatively constrained—yielding a higher modulus and rapid initial relaxation—while rising temperature markedly increases the degrees of freedom of segmental thermal motion, leading to an overall reduction in stiffness and a delayed relaxation process. This observation aligns with the conclusions of He et al. for encapsulation underfills within a modified Qian-Liu framework, namely that mechanical properties deteriorate rapidly near the glass-transition region with a pronounced drop in modulus, and that temperature-dependent apparent activation-energy or hardening-stress terms are required to adjust the predictive model so as to capture this sharp decay.

In the viscoplastic regime, the temperature dependence of the back-stress coefficients  $A$ ,  $A_1$ ,  $A_2$  and the strain-hardening coefficients  $H_1$ ,  $H_2$  further elucidates differences in the inelastic evolution mechanisms. The results show that, with increasing temperature, the  $A$  parameters exhibit a pronounced upward trend, whereas  $A_1$  and  $A_2$  gradually decline, implying that at elevated temperatures the back-stress term contributes more strongly to the magnitude and accumulation rate of inelastic flow, indicative of enhanced viscous drag and dynamic recovery effects. This is consistent with the mechanism emphasized by He et al.<sup>18</sup>, namely increased chain disentanglement and greater freedom of molecular motion. Meanwhile, the hardening coefficients  $H_1$  and  $H_2$  reach a maximum near 60 °C and recede at 85 °C. This “mid-temperature strengthening-high-temperature weakening” behavior accords with the parameter-identification logic of the Qian-Liu framework:  $H_1$  and  $H_2$  are governed by transient features of deformation, and, in the vicinity of the glass-transition region at elevated temperatures, matrix softening reduces the strain-hardening rate, leading to their decline.

It is worth emphasizing that the physical meanings of the back-stress parameters ( $A$ ,  $A_1$ ,  $A_2$ ) and the hardening coefficients ( $H_1$ ,  $H_2$ ) are explicitly defined in the original Qian-Liu framework: the former characterize the accumulation and amplitude modulation of short and long-range back stresses, whereas the latter govern the evolution rate of back stress and the extent of hardening<sup>40</sup>. The temperature-dependent trends identified here are highly consistent with reports of enhanced viscoplastic flow and rapid degradation of mechanical properties at elevated temperatures, thereby further corroborating the applicability of the Qian-Liu model to polymer-based underfill materials.

Taken together, the fitted results not only quantify the temperature-driven evolution of viscoelastic-viscoplastic behavior, but are also mechanistically consistent with the

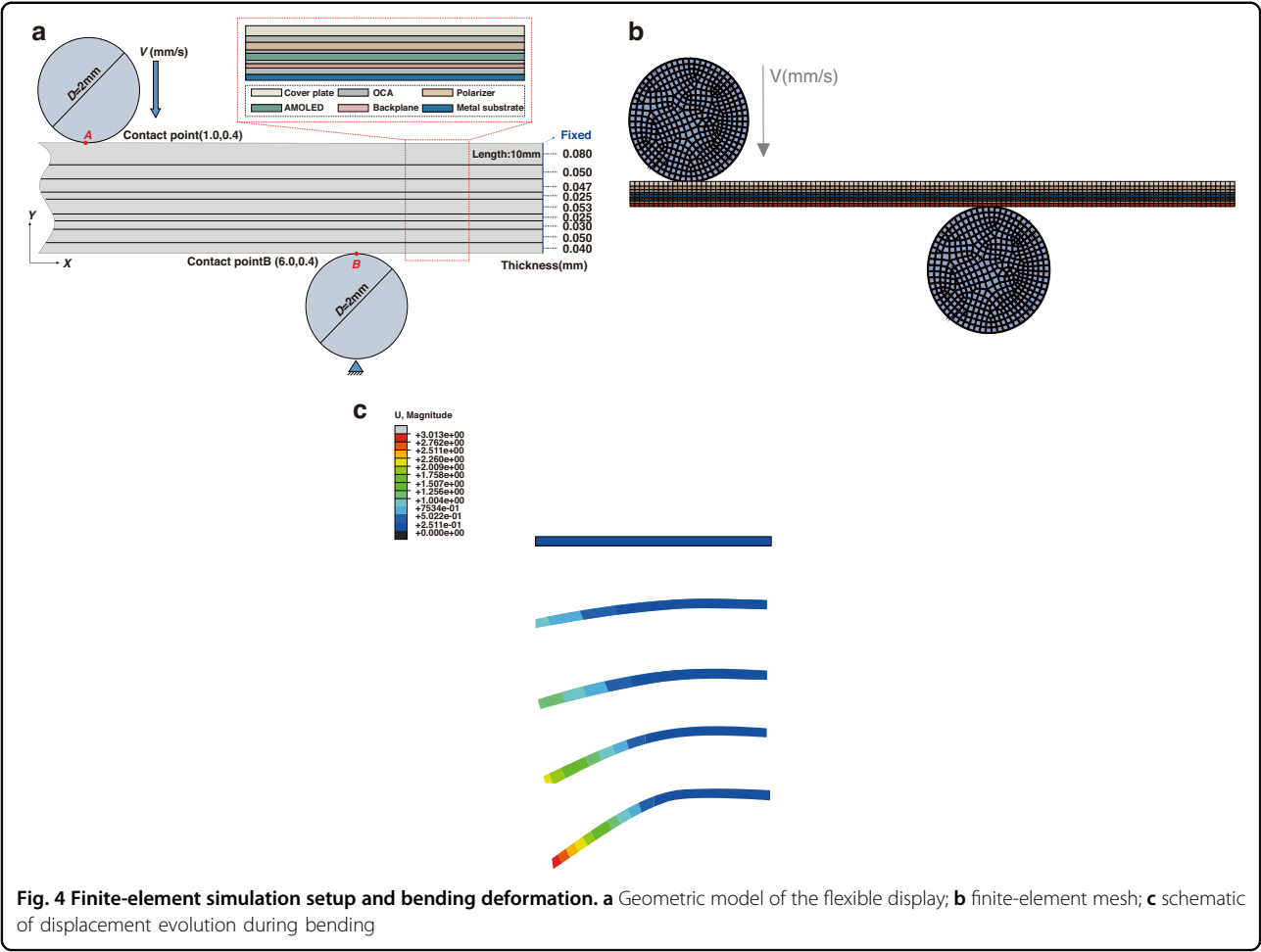
central tenets of the Qian-Liu framework and its subsequent modifications. Parametric analysis confirms that the response is viscoelasticity-dominated at low temperatures, whereas viscoplastic contributions intensify markedly at elevated temperatures; accordingly, back-stress effects and strain-hardening mechanisms exhibit pronounced temperature dependence. These findings provide a robust experimental and theoretical basis for constructing a unified, temperature-dependent constitutive description.

## Finite-Element Simulation Analysis

### Simulation Setup

To investigate the temperature-dependent mechanical response of OCA, a finite-element model of a flexible display was developed in ABAQUS. The overall model represents the thin-film stack of the display, and two rigid bending rollers were introduced at the mid-span to impose a bending configuration, as shown in Fig. 4a. Each roller has a diameter of 2 mm; one roller is fixed while the other translates downward at a prescribed velocity, thereby inducing bending of the adhesive layer and adjacent regions of the display. The right end of the panel is constrained, whereas the remaining portion is allowed to deform freely under bending so as to reflect realistic loading conditions. Contact between the rollers and the display was defined as surface-to-surface hard contact: a “hard” normal constraint was employed, and tangential behavior included friction with a coefficient of 0.02 to better approximate practical conditions. The material parameters assigned to each film layer in the stack are summarized in Table 3, and the main contact settings in ABAQUS are presented in Table 4.

For discretization, the OCA layer was meshed with CPE4 four-node plane-stress elements; the rollers were modeled as rigid bodies, and loading was applied via a reference point. To enhance accuracy, local mesh refinement was introduced in the bending region to capture stress concentrations, as illustrated in Fig. 4b. To ensure numerical accuracy, bending simulations under identical conditions were performed using meshes with 27,489, 43,852, and 160,000 elements. The relative error of the maximum stress among these meshes is less than 5%, confirming that the results are mesh-independent. Regarding material behavior, the functional film layers were treated as linear elastic, whereas the OCA layer was analyzed using two constitutive alternatives: a viscoelastic model based on the generalized Maxwell formulation and a viscoelastic-viscoplastic model that augments the former with a plastic slip mechanism to more faithfully represent irreversible deformation at elevated temperatures. Model parameters were taken from the preceding experimental identification, and simulations were carried out at 30, 60, and 85 °C.



**Fig. 4** Finite-element simulation setup and bending deformation. **a** Geometric model of the flexible display; **b** finite-element mesh; **c** schematic of displacement evolution during bending

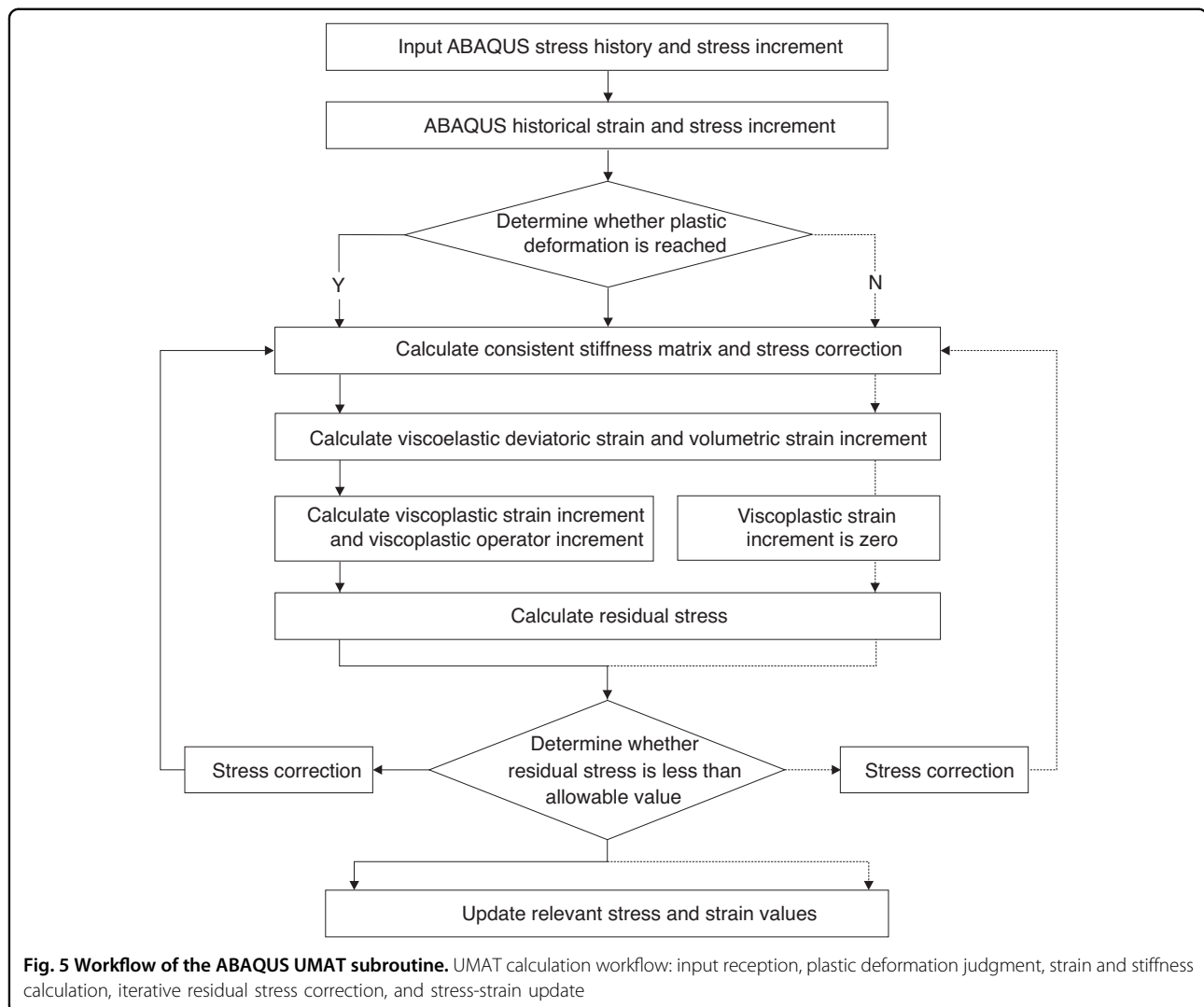
| Table 3 Material parameters of each layer of film |                         |                 |
|---------------------------------------------------|-------------------------|-----------------|
| Film Material                                     | Elasticity Modulus(MPa) | Poisson's ratio |
| Cover Plate                                       | 3500                    | 0.29            |
| Polarizer                                         | 3769                    | 0.33            |
| Backplane                                         | 5030                    | 0.32            |
| Metal Substrate                                   | 56590                   | 0.33            |

| Table 4 Main contact settings in ABAQUS |                        |                    |
|-----------------------------------------|------------------------|--------------------|
| Parameters                              | Details                | Type               |
| Step                                    | Geometric Nonlinearity | on                 |
| Interaction                             | Discretization Method  | Surface to Surface |
|                                         | Normal Behavior        | "Hard" contact     |
|                                         | Tangential Behavior    | Penalty            |
| Load                                    | Boundary Condition     | Fixed/Displacement |
| Geometric Modification                  | Interface Treatment    | Adjust to Touch    |

The evolution of the imposed displacement during bending is depicted in Fig. 4c.

**Simulation Results and Comparative Analysis**

Figure 5 summarizes the UMAT workflow implemented in ABAQUS. Figures 6 and 7 compare the simulated stress fields obtained under identical geometry, loading, and contact conditions using a purely viscoelastic (VE) description and a coupled viscoelastic-viscoplastic (VE-VP) formulation. Both frameworks reproduce the overall topology of the macroscopic stress distribution; however, they differ in peak magnitudes and in their temperature dependence. For the VE model, the peak stress is 58.8 MPa at 30 °C, increasing to 61.1 MPa and 69.1 MPa at 60 °C and 85 °C, respectively, indicating a temperature-induced intensification of stress concentration. In contrast, the VE-VP model predicts consistently lower peak stresses—50.6 MPa at 30 °C, 53.8 MPa at 60 °C, and 54.2 MPa at 85 °C—corresponding to decreases of 13.9%, 11.9%, and 21.6% relative to the VE results. Overall, the VE-VP formulation results in lower peak stresses at all temperatures and a weaker temperature sensitivity



compared with the VE model, indicating moderated stress concentrations under identical geometry, loading, and contact conditions.

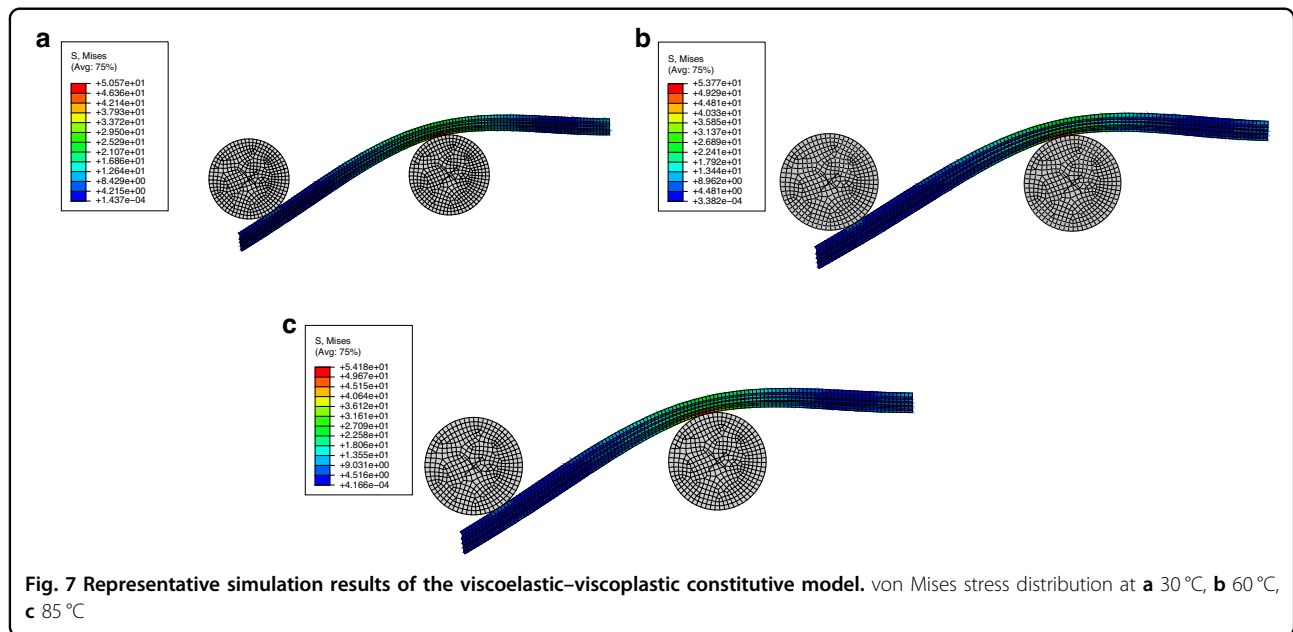
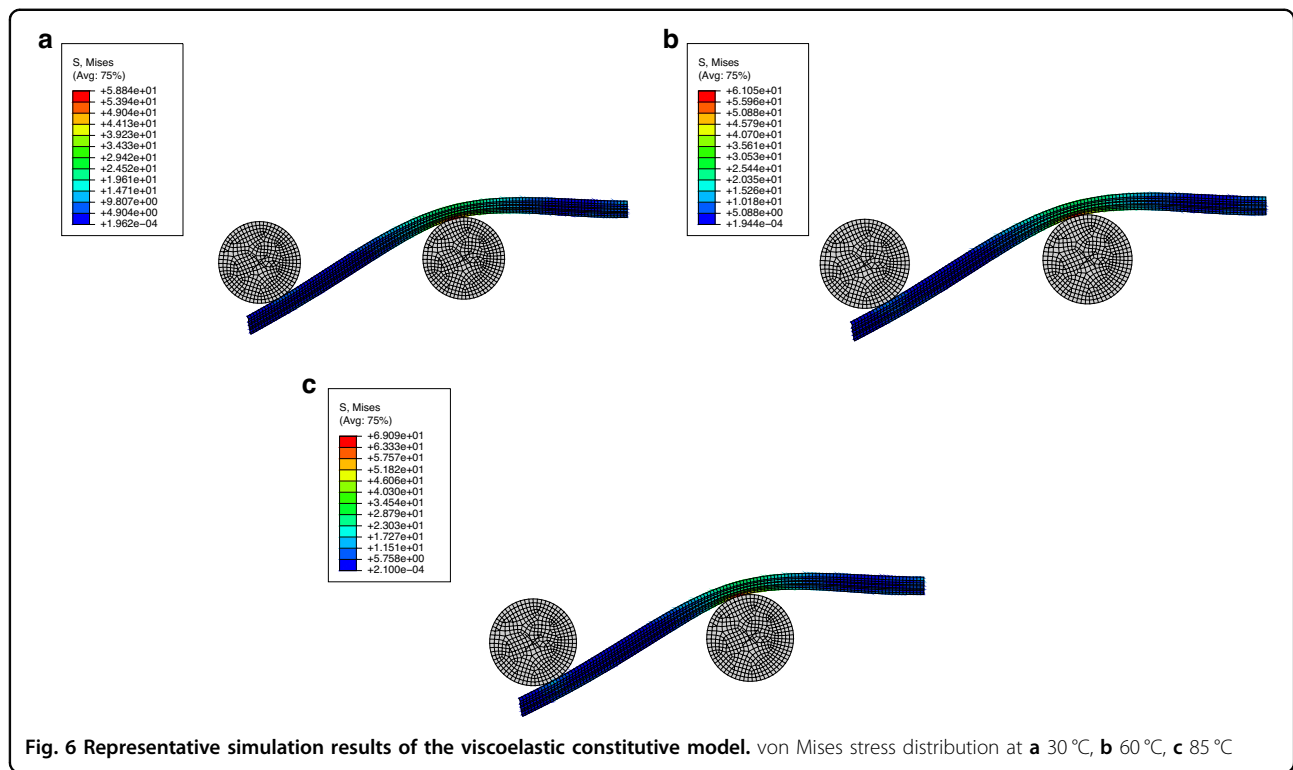
To further elucidate how the viscoelastic and viscoplastic mechanisms give rise to these field-level differences in stress distribution and deformation evolution, we extracted the local stress–strain response and the strain–energy evolution from ABAQUS. The corresponding output variables (stress–strain and strain energy–time) are summarized in Table 5. The extracted data were then used to plot the stress–strain and strain energy–time curves for both the VE and VE-VP models at different temperatures, as shown in Fig. 8. The results show that, at the initial stage, the local strain levels predicted by the two models remain low and their responses are relatively close. As bending proceeds into the intermediate-to-late stage, the discrepancy between the two frameworks rapidly amplifies: the coupled VE-VP formulation

markedly suppresses the local stress level compared with the purely viscoelastic description, consistent with the reduced peak stresses observed in Figs. 6 and 7.

From an energy perspective, Table 5 further indicates that, under the same bending conditions, the accumulated strain energy (ALLSE) predicted by the VE-VP model is consistently and significantly lower than that of the VE model at all temperatures. This demonstrates that introducing viscoplasticity reduces the recoverable energy storage capacity of the system and effectively constrains the overall energy accumulation, thereby providing an energetic explanation for the moderated stress concentration and weakened temperature sensitivity captured by the VE-VP framework.

## Conclusions

This study investigates the mechanical response of OCA across multiple temperatures and systematically develops



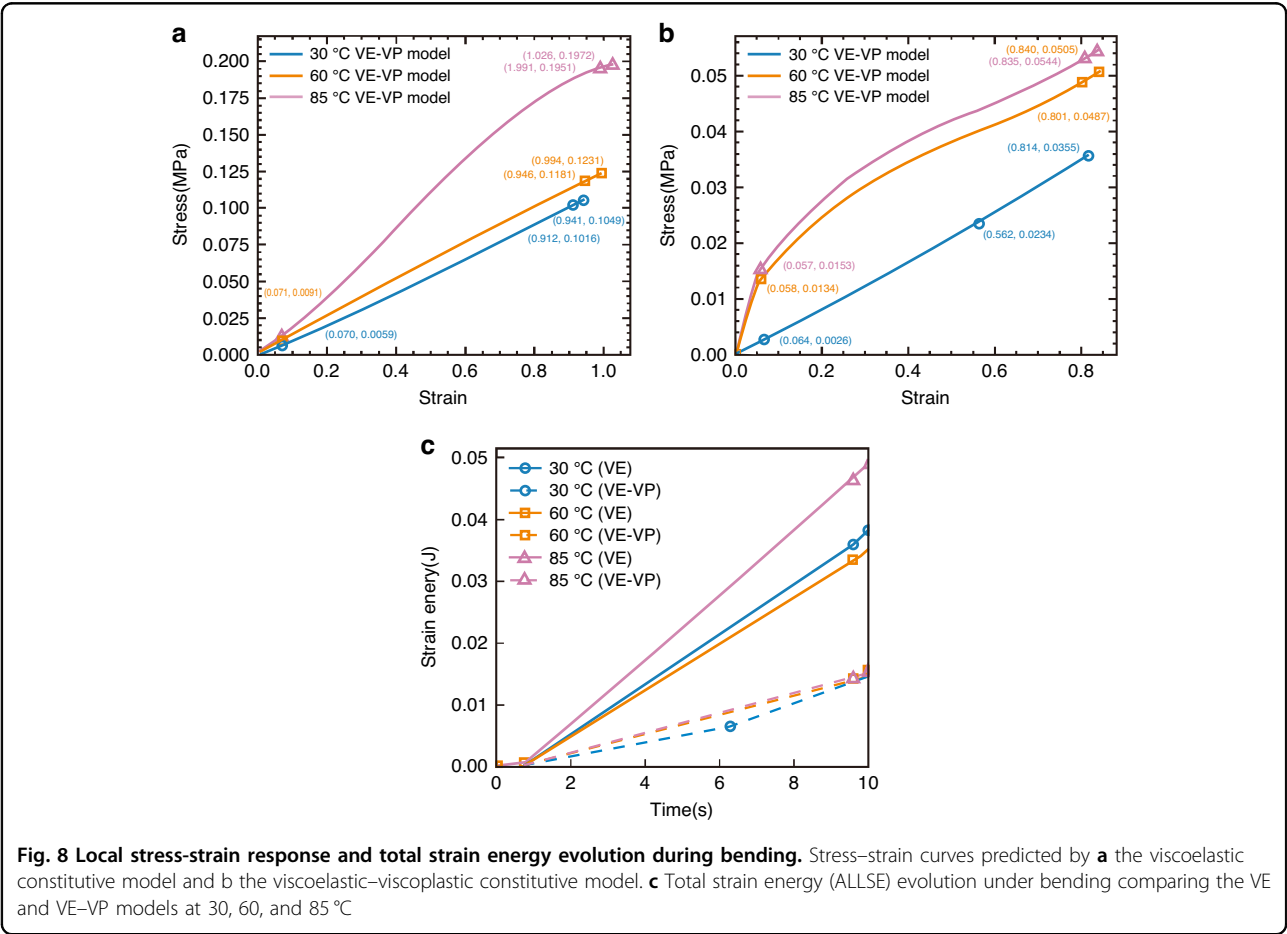
and compares two constitutive descriptions—viscoelastic and viscoelastic-viscoplastic. Key parameters were identified from stress-relaxation and creep-recovery experiments and validated via finite-element simulations. The principal conclusions are as follows:

- (1) The viscoelastic mechanical response was characterized using a generalized Maxwell model,

revealing a pronounced temperature dependence: increasing temperature enhances segmental mobility and reduces internal dissipation, thereby decreasing both the equilibrium stress and the equilibrium relaxation modulus. At 85 °C the initial stress is approximately 12.6 kPa and the equilibrium relaxation modulus is about 9.51 kPa,

**Table 5** Key response indicators from FE simulations for VE and VE-VP models

| Temperature | viscoelastic constitutive model |        |          |               | viscoelastic–viscoplastic constitutive model |        |         |               |
|-------------|---------------------------------|--------|----------|---------------|----------------------------------------------|--------|---------|---------------|
|             | Stress (MPa)                    | Strain | Time (s) | Strain Energy | Stress(MPa)                                  | Strain | Time(s) | Strain Energy |
| 30°C        | 0.0058                          | 0.069  | 0.758    | 0.00030       | 0.0026                                       | 0.064  | 0.758   | 9.19e-5       |
|             | 0.102                           | 0.912  | 6.28     | 0.0359        | 0.0234                                       | 0.562  | 6.28    | 0.0066        |
|             | 0.105                           | 0.941  | 10       | 0.0382        | 0.0355                                       | 0.814  | 10      | 2.0149        |
| 60°C        | 0.0091                          | 0.071  | 0.758    | 0.00041       | 0.0134                                       | 0.058  | 0.758   | 0.00023       |
|             | 0.118                           | 0.946  | 6.28     | 0.0334        | 0.0487                                       | 0.800  | 6.28    | 0.0142        |
|             | 0.123                           | 0.994  | 10       | 0.0353        | 0.0505                                       | 0.840  | 10      | 0.0155        |
| 85°C        | 0.0125                          | 0.069  | 0.758    | 0.00051       | 0.0153                                       | 0.057  | 0.758   | 0.00026       |
|             | 0.195                           | 0.991  | 6.28     | 0.0463        | 0.0531                                       | 0.807  | 6.28    | 0.0145        |
|             | 0.197                           | 0.995  | 10       | 0.0491        | 0.0544                                       | 0.835  | 10      | 0.0158        |



**Fig. 8** Local stress-strain response and total strain energy evolution during bending. Stress–strain curves predicted by **a** the viscoelastic constitutive model and **b** the viscoelastic–viscoplastic constitutive model. **c** Total strain energy (ALLSE) evolution under bending comparing the VE and VE–VP models at 30, 60, and 85 °C

representing reductions of 37.72% and 23.61% relative to those at 30 °C and 60 °C, respectively. In dynamic mechanical tests, the storage modulus decreases monotonically with increasing scan temperature—from 1666.3 MPa down to

0.6628 MPa—with the steepest decline occurring near the glass-transition temperature( $T_g = 0\text{ }^{\circ}\text{C}$ ). A generalized Maxwell model with a three-term Prony series accurately reproduces the master-curve data, and temperature shift factors in both

the high and low-temperature regimes are well captured by the corresponding fits.

- (2) In parallel, a viscoelastic-viscoplastic model, constructed by coupling the generalized Maxwell model with the Qian-Liu framework, was compared with the proposed viscoelastic model. Both models faithfully capture the salient features of the OCA stress-relaxation data and accurately reflect the temperature dependence of its mechanical behavior. During the parameter fitting process using the viscoelastic-viscoplastic model, the viscoplastic-related parameters at 60 °C and 85 °C significantly exceed those at 30 °C. This phenomenon indicates that the viscoelastic characteristics of the OCA adhesive diminish with increasing temperature.
- (3) ABAQUS-based simulations furnish robust theoretical and numerical support for the structural optimization and reliability assessment of flexible OLED devices.

The present results demonstrate that the proposed constitutive-FE framework can effectively capture the temperature-dependent mechanical response of the OCA and provides a feasible basis for device-level stress assessment. However, the current implementation remains subject to limitations: the material parameters were calibrated for a specific OCA under a restricted set of experimental conditions, and the finite-element analysis was conducted using a simplified two-dimensional configuration. Therefore, further work is warranted to validate and extend the framework under more complex loading paths and within more realistic three-dimensional structural models.

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#### Data availability

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

#### Competing interests

Sheng Liu is an Editor for the journal, the remaining authors have no competing interest.

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